The Management University of Africa

MUA

Sponsored by the Kenya Institute of Management

BDS 109 WBA **ANSWER ALL QUESTIONS**

1. A machine comprises of 3 transformers A, B and C. The machine may operate if at least 2 transformers are working. The probability of each transformer working are given as shown below;

   P (A) = 0.6, P (B) = 0.5, P(C) = 0.7

   A mechanical engineer went to inspect the working conditions of those transformers. Find the probabilities of having the following outcomes

   a. Only one transformer operating **(2 marks)**

   b. Two transformers are operating **(2 marks)**

   c. All three transformers are operating **(1 marks)**

2. The following table indicates the marks obtained by students in a statistics test.

Marks	Number of students
0 – 20	5
20 – 40	7
40 – 60	-
60 – 80	8
80 – 100	7

The arithmetic mean for the class was 52.5 marks. You are required to determine the value of:

i) The missing frequency **(5 marks)**

ii) The median mark **(2 marks)**

iii) The modal mark **(2 marks)**

iv) The standard deviation **(4 marks)**

v) The coefficient of skewness **(2 marks)**